Ain Shams University

Faculty of Computer and Information Sciences (FCIS)

Task manager

Academic Year: 2025 / 2026

Team Members

#	Full Name	Student ID	Phone
1	Ahmed Omar	2023170809	01211239624
2	Mohamed Mostafa	20241701136	01550011688
3	Sufyan Ahmed Hassan	2023170789	01557410621
4	Mohamed Seif Saeed	2023170808	01155683419
5	Ahmed Medhat Mostafa Sayed Mohamed	2023170063	01023087511
6	Ahmed Hisham Mohamed Awad	20191700079	01141573566
	Section : SC_6/4		**19/8/2026**

Student Task & Study Planner

1. Project Overview

The **Student Task & Study Planner** is an Android application developed to assist students in organizing, managing, and monitoring their academic tasks. The application provides a simple and intuitive interface through which users can create, view, update, delete, and track their study-related tasks.

The application uses a local **SQLite database** to store task information. This approach allows the application to operate without an internet connection while maintaining persistent data on the user's device.

The project focuses on three main aspects:

1. Effective task management through CRUD operations.
2. Persistent offline data storage using SQLite.
3. Statistical analysis of task progress to help students monitor their academic workload.

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2. Project Objectives

The main objective of the project is to develop a lightweight and reliable task-management application specifically designed for students.

The project aims to:

- Provide an easy-to-use interface for managing academic tasks.
- Allow users to create and organize tasks according to their subjects.
- Enable users to assign priorities to tasks.
- Track the progress of each task.
- Provide complete CRUD functionality.
- Store task information persistently using SQLite.
- Allow the application to function without an internet connection.
- Provide statistical information about task completion and progress.

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3. System Functionalities

The application provides several functionalities that cover the complete task-management process.

3.1 Main Dashboard and Navigation

The `MainActivity` serves as the main navigation hub of the application.

From the main interface, users can access the following sections:

- Add Task
- View Tasks
- Statistics

The activity also contains an Options Menu implemented using `main_menu.xml`. The menu provides an About/Information option that displays application information using an Android Toast message.

3.2 Task Creation

The `AddTaskActivity` is responsible for creating new tasks.

The task creation form allows the user to enter the following information:

Attribute	Description
Title	The name of the academic task
Description	Detailed information about the task
Subject	The academic subject associated with the task
Priority	The priority level of the task
Status	The current progress of the task

The priority can be selected from three options:

- Low
- Medium
- High

The task status can also be selected from:

- Not Started
- In Progress
- Completed

Form Validation

Before a task is stored in the database, the application validates the required input fields. For example, the task title must not be empty.

If the required information is missing, the application prevents the insertion and provides appropriate feedback to the user.

Database Insertion

After successful validation, the task information is inserted into the SQLite database through the `TABLE_TASKS` table.

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4. Task Management

4.1 Task List

The `TaskListActivity` is responsible for displaying all stored tasks.

Tasks are presented dynamically using an Android `RecyclerView`. Each individual task is represented using the custom `task_row.xml` layout.

The RecyclerView allows the application to efficiently display multiple tasks while maintaining a structured and consistent user interface.

4.2 Task Interaction

The application provides multiple methods for interacting with individual tasks.

Normal Click

When a user clicks on a task, the application opens the task details screen. The existing task information is displayed and can be modified.

Long Click

A long click on a task opens an Android Context Menu containing actions such as:

- Edit
- Delete

Popup Menu

Each task row also provides a three-dot Popup Menu. This menu allows the user to:

- Edit the selected task.
- Delete the selected task.

When a task is deleted, the RecyclerView is refreshed so that the interface immediately reflects the updated database contents.

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5. Task Updating and Deletion

5.1 Task Details

The `TaskDetailsActivity` is responsible for displaying and managing an existing task.

When a task is selected, its `task_id` is passed to the activity through an Android `Intent`.

The application uses this ID to retrieve the corresponding record from the SQLite database and populate the appropriate input fields.

This allows the user to review and modify the existing task information.

5.2 Task Updating

Users can modify the following task attributes:

- Title
- Description
- Subject
- Priority
- Status

After the user saves the changes, the updated information is stored in the SQLite database.

A Toast notification is used to provide feedback regarding the successful update operation.

5.3 Task Deletion

The application also allows users to permanently delete tasks from the database.

Deletion can be performed through the task details interface or through the available task menus.

After deletion, the user receives feedback through a Toast notification, and the task list is updated accordingly.

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6. Statistics and Progress Tracking

6.1 Statistics Activity

The `StatisticsActivity` provides users with an overview of their current academic tasks.

The activity queries the SQLite database and calculates the number of tasks belonging to each status category.

The following statistics are displayed:

- Total number of tasks
- Number of completed tasks
- Number of tasks currently in progress
- Number of tasks that have not been started

For example:

Status	Number of Tasks
Completed	8
In Progress	7
Not Started	5
Total	**20**

This functionality allows users to quickly evaluate their current workload and monitor their progress.

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7. Database Design

The application uses **SQLite** as its local database management system.

Database creation and management are handled using the Android `SQLiteOpenHelper` class.

The main table used by the application is:

`TABLE_TASKS`

The database supports all four fundamental CRUD operations:

Operation	Description
Create	Adds a new task to the database
Read	Retrieves and displays stored tasks
Update	Modifies an existing task
Delete	Removes a task from the database

Using SQLite provides persistent local storage, allowing task information to remain available after the application is closed or restarted.

8. Application Components

The main components of the application are summarized below:

Component	Responsibility
`MainActivity`	Provides the main dashboard and navigation
`AddTaskActivity`	Handles creation of new tasks
`TaskListActivity`	Displays the stored tasks
`TaskDetailsActivity`	Displays, updates, and deletes existing tasks
`StatisticsActivity`	Calculates and displays task statistics
`TaskAdapter`	Connects task data with the RecyclerView
`SQLiteOpenHelper`	Creates and manages the SQLite database
`main_menu.xml`	Defines the application's main Options Menu
`task_row.xml`	Defines the layout of individual task rows

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9. Technologies Used

The following technologies and Android components were used in the development of the application:

- **Android**
- **Java**
- **SQLite**
- **SQLiteOpenHelper**
- **RecyclerView**
- **Android Intents**
- **Options Menu**
- **Context Menu**
- **Popup Menu**
- **Toast Notifications**
- **XML Layouts**

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10. Offline Data Management

A major feature of the application is its ability to operate completely offline.

Unlike applications that depend on a remote server or cloud database, the Student Task & Study Planner stores all task information locally on the Android device.

The general data flow is:

User → Android Interface → Application Logic → SQLite Database → Local Device Storage

This approach eliminates the requirement for an internet connection and allows users to access and manage their tasks at any time.

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11. User Interface and User Experience

The application was designed with simplicity and ease of use in mind.

The interface provides:

- Clear navigation between application sections.
- Structured task presentation.
- Simple forms for entering task information.
- Multiple methods for editing and deleting tasks.
- Immediate feedback through Toast notifications.
- A dedicated statistics interface for monitoring task progress.

The use of standard Android components such as RecyclerView, Menus, Spinners, RadioGroups, and Intents provides a familiar interaction model for Android users.

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12. Future Improvements

Although the current application provides the essential functionality required for academic task management, several improvements could be implemented in future versions.

Possible improvements include:

- Task reminders and scheduled notifications.
- Calendar-based task management.
- Search and filtering functionality.
- Task categories and tags.
- More advanced statistical analysis.
- Dark mode support.
- Cloud synchronization.
- User authentication.
- Task backup and data export.
- Deadline tracking and overdue-task notifications.

13. Conclusion

The **Student Task & Study Planner** successfully provides a lightweight Android-based solution for managing academic tasks.

The application implements complete CRUD functionality, allowing users to create, retrieve, update, and delete tasks. SQLite provides reliable local data persistence, while RecyclerView provides an efficient mechanism for displaying task information.

In addition, the Statistics functionality provides users with a clear overview of their task progress by categorizing tasks into Completed, In Progress, and Not Started states.

Overall, the project demonstrates the practical implementation of Android application development concepts, including user interface design, activity navigation, form validation, RecyclerView usage, SQLite database management, and data analysis.

- **Interactive Menus:**
  - **Normal Click:** Opens full task details for editing.
  - **Long Click (Context Menu):** Triggers an Android Context Menu allowing fast Edit and Delete actions.
  - **Row Menu (Popup Menu):** Triggers an overflow menu (three-dot menu) per list item to perform Edit or direct SQLite Deletion with auto-refreshing UI.

D. Task Updating & Deletion (`TaskDetailsActivity`)

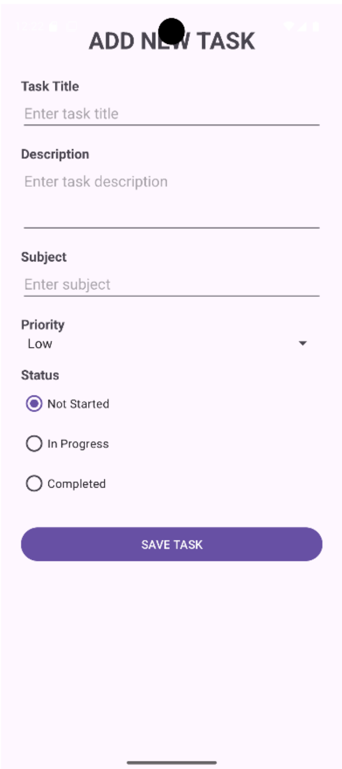
- **Pre-populated Fields:** Fetches existing task records using `task_id` extras passed through Android `Intent` and fills input controls dynamically.
- **Database Update & Delete Operations:** Allows users to modify details or remove records from SQLite with feedback provided via `Toast` notifications.

E. Real-time Progress Tracking (`StatisticsActivity`)

- **Task Analytics:** Dynamically queries the SQLite database to compute and display total task counts and categorizes them by status (*Completed*, *In Progress*, and *Not Started*).

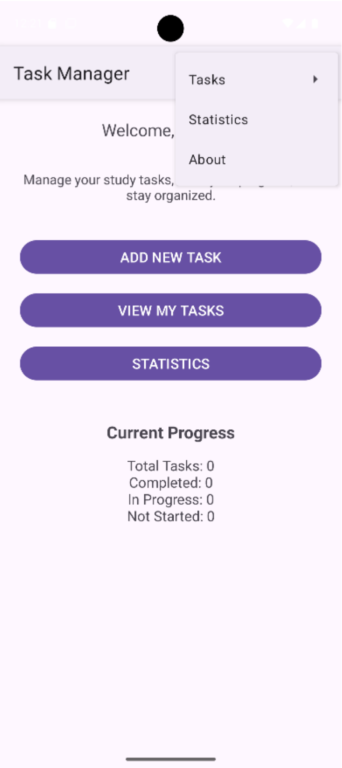
Screenshot 1: Main Dashboard (MainActivity)

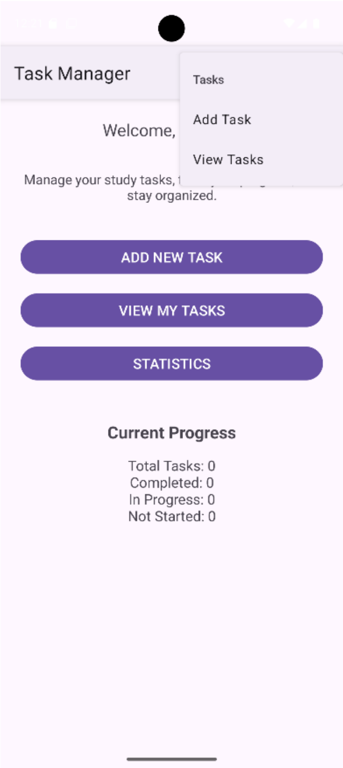
Screenshot 2: Add Task Form (AddTaskActivity)



Screenshot 3: Task List View (TaskListActivity)

Screenshot 4: Sub & Popup Menus





Screenshot 5: Statistics Screen (StatisticsActivity)